Chapter 13: Conclusions and Future Developments

Nature Inspired Optimisation for Delivery Problems (2022)

Based on: T. Cherrett, J. Holguin-Veras *et al.*

2022 — Slides prepared for self-study

Chapter 13 — Overview I

This final chapter of the book takes stock of everything covered in the preceding twelve chapters and looks ahead to three major areas where the field of delivery optimisation is expected to evolve rapidly.

Chapter Structure

- **Section 13.1 — Software, Algorithms and AI.** How the next generation of optimisation *software* will be designed, focusing on the concept of a *domain-specific AI solver* that can autonomously understand and solve a broad class of delivery problems.
- **Section 13.2 — Developments in Transportation.** Emerging physical technologies — autonomous vehicles, drones, electric fleets, and the use of public transport as a freight carrier — that will reshape the landscape in which optimisation algorithms must operate.
- **Section 13.3 — Consumers.** How changing consumer expectations (speed, sustainability, transparency) and growing awareness of climate change are creating new requirements for delivery systems.

Chapter 13 — Overview II

Why This Chapter Matters

Throughout this book, the authors have shown that nature-inspired algorithms (evolutionary algorithms, ant-colony optimisation, particle-swarm optimisation, etc.) can solve difficult routing problems. This concluding chapter asks: *where does the field go from here?* The answer lies at the intersection of better AI, better physical transport technology, and better understanding of what consumers actually want.

13.1 Software, Algorithms and AI — The Challenge I

One of the most practically important questions in delivery logistics is deceptively simple: *how should a company choose the right optimisation software for their specific problem?*

The difficulty is that real delivery problems come in an enormous variety of forms. A parcel carrier faces a very different problem from a food-delivery platform or a humanitarian aid organisation, yet all of them share a common structure: vehicles, customers, routes, and objectives such as cost or time.

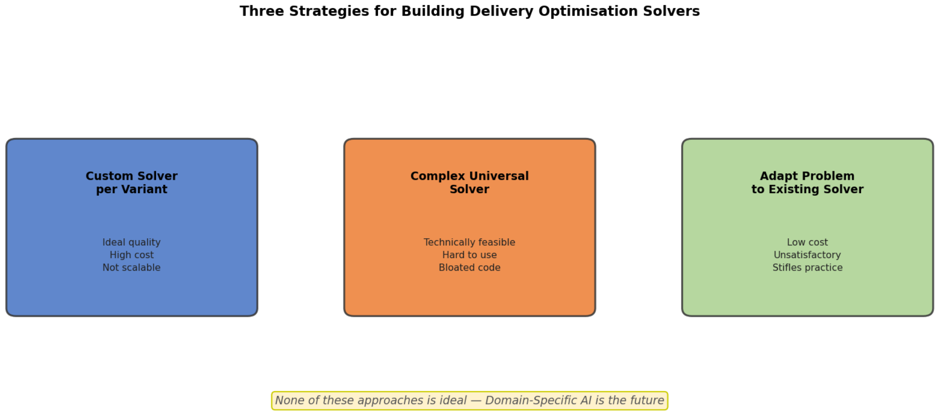
13.1 Software, Algorithms and AI — The Challenge II

Three Existing Strategies — and Their Weaknesses

The book identifies three strategies currently used to match problems to solvers, and explains why none of them is fully satisfactory:

1. **Build a custom solver for each variant.** This produces the best possible solution quality for a given problem, but the cost is enormous: a specialist team must be assembled for every new variant (time windows, multiple depots, heterogeneous fleets, etc.), and the resulting code is rarely reusable.
2. **Build one enormous universal solver.** Such a solver would cover all variants in a single codebase. While technically conceivable, the result is an unwieldy, hard-to-maintain system that is difficult for practitioners to understand and configure correctly.
3. **Force the problem to fit an existing solver.** This is the most common approach in practice: simplify or distort the real problem so that an off-the-shelf package can handle it. The cost is solution quality — the algorithm optimises the *wrong* objective or ignores important constraints.

13.1 Software, Algorithms and AI — The Challenge III



13.1 Software, Algorithms and AI — The Challenge IV

Key Insight

All three strategies represent a compromise. The ideal solution would be software that *understands* the problem deeply — much as a human expert does — and constructs the right algorithm automatically. This is the vision for the next generation of AI-driven optimisation tools.

13.1 The Vision: An Adaptive Problem Solver I

The book proposes a more ambitious goal: building an **adaptive problem solver** that sits somewhere between a rigid specialist tool and the far-future concept of Artificial General Intelligence (AGI).

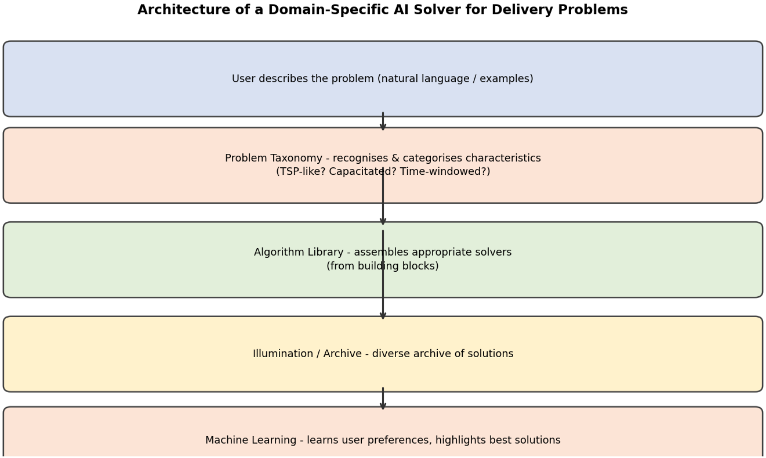
The key idea is that the solver should be able to do the following things automatically, without requiring manual reconfiguration for each new problem instance:

13.1 The Vision: An Adaptive Problem Solver II

Capabilities of an Adaptive Solver

- **Understand the problem from a description.** Given a plain-language or structured description of a new delivery scenario, the solver recognises which class of problem it belongs to (TSP-like? capacitated VRP? time-windowed? multi-depot?).
- **Select and configure appropriate algorithms.** Rather than applying a fixed metaheuristic, the solver draws on a library of algorithmic building blocks — local-search operators, crossover strategies, population-management schemes — and assembles the combination most likely to perform well for this problem class.
- **Learn from experience.** Machine learning techniques allow the solver to track which algorithm configurations have worked well on similar problems in the past, and to give preference to those configurations when encountering new but structurally similar instances.
- **Present a diverse set of solutions.** Using an *illumination* approach (such as MAP-Elites, covered in Chapter 6), the solver maintains a rich archive of solutions that trade off different objectives (e.g., cost vs. carbon emissions vs. delivery time), so that decision-makers can explore the trade-off space.

13.1 The Vision: An Adaptive Problem Solver III



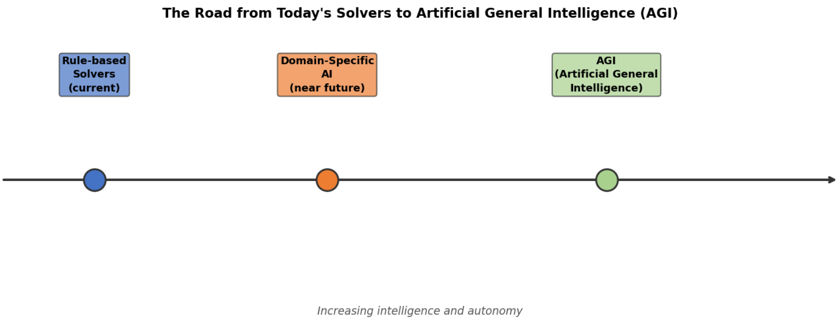
13.1 The Vision: An Adaptive Problem Solver IV

How to Read This Diagram

The architecture flows *downward*. A user provides a problem description at the top; the taxonomy layer classifies it; the algorithm library assembles a solver; the illumination layer generates a diverse solution archive; and the machine-learning layer surfaces the solutions that best match the user's implicit preferences (for example, preferring greener routes even if the user has not explicitly requested this).

13.1 The Road to AGI — A Broader Perspective I

The adaptive solver described above is a near-term, *domain-specific* AI system: it is highly capable within delivery logistics, but it is not a general-purpose reasoner. The book situates this in a longer trajectory:



13.1 The Road to AGI — A Broader Perspective II

Three Stages in the Evolution of AI for Logistics

Rule-based solvers (today). Current commercial software applies a fixed set of rules and a single metaheuristic. The user must manually select the algorithm and tune its parameters. Solutions are good but the system has no understanding of the problem — it simply executes a procedure.

Domain-specific AI (near future). Software that understands problem structure, assembles algorithms automatically, and learns from previous runs. This is the target described in Section 13.1.

Artificial General Intelligence (AGI). A system with human-level reasoning across arbitrary domains. AGI could, in principle, solve *any* combinatorial optimisation problem after reading a description. Futurist Ray Kurzweil (cited in the book) argued that human-level AI could arrive as early as 2029, though this remains a subject of active debate.

13.1 The Road to AGI — A Broader Perspective III

Practical Takeaway

Whether or not AGI arrives by 2029, the authors argue that *the development of domain-specific AI solvers for logistics is already underway and will fundamentally change how companies deploy optimisation software.* The techniques in this book — illumination, multi-objective search, real-world data integration — are the building blocks of that future.

Python: Skeleton of an Adaptive Solver

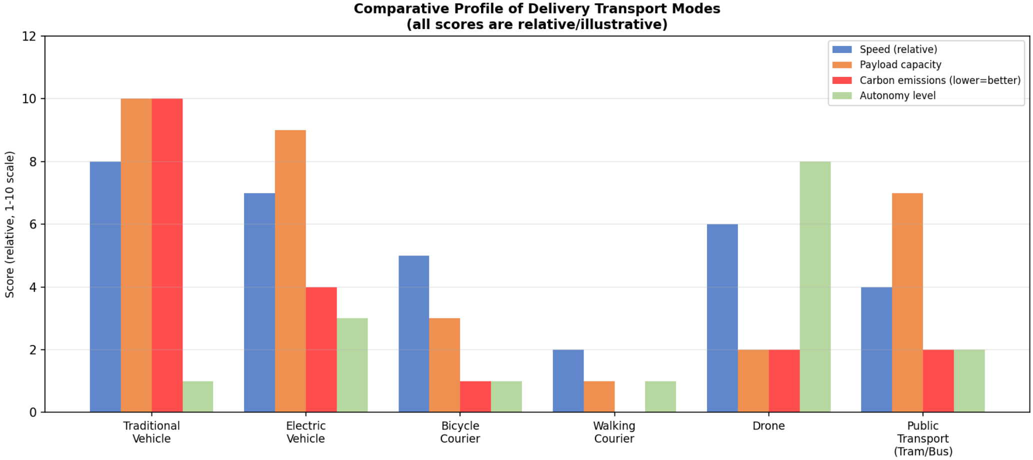
The following Python sketch illustrates how an adaptive solver might be structured in practice. The core idea is that the `solve()` function does not hard-code a specific algorithm; instead it *selects* the best available algorithm based on the problem's characteristics.

```
class AdaptiveSolver:
    """
    Domain-specific AI solver for delivery problems.
    Automatically selects and configures the best algorithm
    based on problem characteristics detected at runtime.
    """

    def __init__(self, algorithm_library):
        # algorithm_library is a dict: name -> algorithm class
        self.library = algorithm_library
        self.performance_log = {}   # tracks past performance

    def classify_problem(self, problem):
        """
        Inspect the problem and return a dict of characteristics.
        E.g.: {'type': 'CVRP', 'n_customers': 200,
               'has_time_windows': True, 'multi_depot': False}
        """
        characteristics = {}
        characteristics['n_customers'] = len(problem.customers)
        characteristics['has_time_windows'] = problem.time_windows is not None
        characteristics['has_capacity']     = problem.vehicle_capacity is not None
        characteristics['multi_depot']      = len(problem.depots) > 1
```

13.2 Developments in Transportation — Overview I

Even the most sophisticated algorithm can only work with the physical transport infrastructure available to it. Section 13.2 examines how the physical side of delivery is changing, and why these changes create new and harder optimisation problems.

13.2 Developments in Transportation — Overview II



13.2 Developments in Transportation — Overview III

Four Major Technological Shifts

1. **Autonomous vehicles** — removing the human driver from the cost function.
2. **Unmanned Aerial Vehicles (UAVs / drones)** — accessing locations unreachable by road at very low cost per delivery.
3. **Electric and alternative-fuel vehicles** — changing the cost structure (energy vs. fuel) and range constraints of the fleet.
4. **Public transport as freight carrier** — using existing tram, bus, and rail networks to move parcels to urban micro-depots.

13.2 Autonomous Vehicles I

An autonomous vehicle (AV) is one that navigates and drives without a human operator. The book notes that AVs offer two distinct advantages for delivery logistics:

Advantages of Autonomous Delivery Vehicles

- **Cost reduction.** The single largest cost in last-mile delivery is the *driver's wage*. An autonomous vehicle eliminates this cost entirely, potentially making small-load deliveries economically viable for the first time.
- **Flexibility.** An AV can operate 24 hours a day without fatigue regulations, enabling overnight or off-peak deliveries that reduce urban congestion during rush hours.

The book also highlights a key research challenge that emerges with AVs: **the multi-agent coordination problem**. When many AVs from different companies operate in the same city, they must cooperate to avoid collisions, share charging infrastructure, and prevent the streets from becoming congested with empty-running vehicles.

13.2 Autonomous Vehicles II

Worked Example: Multi-Agent Savings

Suppose two delivery companies each operate 10 AVs in the same city. Without coordination, both fleets may converge on the same streets at the same time, doubling congestion and halving the effective road capacity. With a shared routing protocol (e.g., a city-level swarm coordinator), the 20 AVs can be dispatched on non-overlapping routes, reducing total distance driven by an estimated 15–30% (a figure consistent with cooperative VRP studies in the literature).

Key Insight

The introduction of AVs does not simplify the optimisation problem — it transforms it from a *single-company* Vehicle Routing Problem into a *multi-agent* coordination problem that may require city-level governance as well as algorithmic innovation.

13.2 Drones and Unmanned Aerial Vehicles (UAVs) I

Unmanned Aerial Vehicles (UAVs), commonly called drones, can fly directly from a depot or a truck to a customer's door, bypassing road congestion entirely. The book cites multiple studies on drone-based delivery and identifies both the promise and the constraints:

Why Drones Are Attractive for Last-Mile Delivery

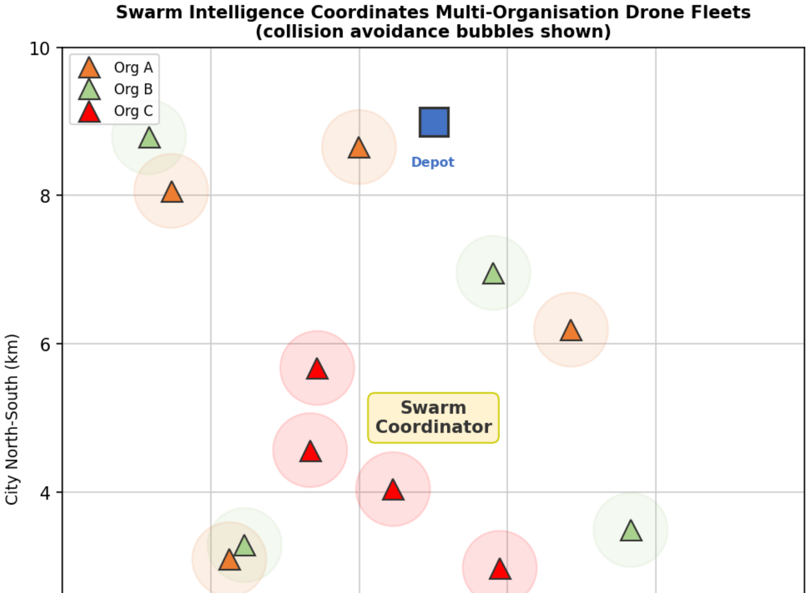
- **Speed.** A drone travels in a straight line ("as the crow flies"), avoiding road distance that can be 40–60% longer than the geodesic distance in urban grids.
- **Access.** Rural or island locations that are expensive to serve by road can be reached at low marginal cost. The book cites drone delivery in the Pacific Islands and healthcare supply to remote clinics as real-world examples.
- **Low carbon footprint per delivery.** An electric drone carrying a small parcel produces far less CO_2 per delivery than a conventional van on a half-empty route.

13.2 Drones and Unmanned Aerial Vehicles (UAVs) II

Constraints and Research Challenges

- **Payload limit.** Current commercial drones carry at most 2–5 kg, which covers roughly 80% of e-commerce parcels but excludes bulky items.
- **Range and battery.** Most delivery drones have an operational range of 10–20 km per charge. This means they must be recharged or replaced frequently, and the placement of charging stations becomes an important optimisation sub-problem.
- **Regulation.** Most national aviation authorities require drones to remain within visual line of sight (VLOS), which limits autonomous long-range operation. The book notes that regulation is evolving rapidly.
- **Weather sensitivity.** Strong winds, rain, and fog can ground drone fleets, introducing uncertainty that must be handled by the routing algorithm.
- **Multi-organisation coordination.** When multiple companies operate drone fleets over the same city, a swarm-intelligence layer is needed to prevent mid-air collisions and to allocate airspace fairly.

13.2 Drones and Unmanned Aerial Vehicles (UAVs) III



13.2 Electric and Alternative-Fuel Vehicles I

Electric vehicles (EVs) are already being adopted by major parcel carriers in Europe and North America. Their impact on routing optimisation is not just environmental — it changes the *mathematical structure* of the problem.

13.2 Electric and Alternative-Fuel Vehicles II

How EVs Change the VRP

A conventional vehicle routing problem assumes that a vehicle can travel any distance as long as it does not exceed its *load capacity*. An electric vehicle adds a second constraint: the *state of charge* (SoC), which decreases with distance and increases only when the vehicle visits a charging station. This gives rise to the **Electric Vehicle Routing Problem (E-VRP)**, which has the following additional features:

- **Charging station placement.** The locations of charging stations must be known in advance, and routes must be designed to visit them before the battery runs empty.
- **Non-linear charging time.** Batteries charge quickly when nearly empty but slowly when nearly full (the standard lithium-ion charge curve). This means a full recharge is rarely optimal: partial recharges at multiple stops may minimise total route time.
- **Range anxiety.** Even with careful planning, unexpected traffic or detours can drain the battery unexpectedly. Robust routing algorithms must plan with a safety margin.

13.2 Electric and Alternative-Fuel Vehicles III

Worked Example: Range Constraint in a Route

Suppose an EV has a range of 120 km on a full charge and must serve five customers at distances (from depot) of 15, 30, 55, 85, and 110 km. A naive nearest-neighbour route visits them in order: $0 \to 15 \to 30 \to 55 \to 85 \to 110 \to 0$ km, for a total of approximately 140 km — *exceeding* the 120 km range. The vehicle would run out of charge before returning to the depot. A good E-VRP solver would insert a stop at a charging station located, say, at km 70, adding perhaps 20 minutes of charging time but ensuring the route is feasible. The trade-off between route length and charging stops is the core optimisation challenge.

The book also notes that alternative fuels — hydrogen, compressed natural gas, and hybrid propulsion — are entering the fleet mix, each with different range, refuelling infrastructure, and environmental profiles. Future solvers must be able to handle *heterogeneous* fleets containing vehicles with different fuel types simultaneously.

13.2 Public Transport as a Freight Carrier I

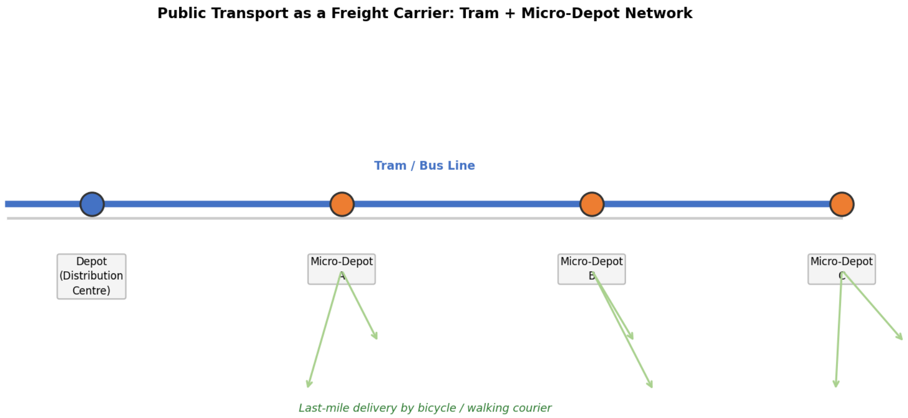
An innovative idea gaining traction in European cities is to use **existing public transport networks** — trams, buses, underground trains — to carry parcels alongside passengers.

The concept addresses a fundamental problem in urban logistics: delivery vans travelling through city centres cause congestion and pollution precisely at the times and places where pedestrians and cyclists are most present. If parcels could instead travel on trams that are already running, the number of delivery vans in the city centre could be reduced dramatically.

How a Tram-Based Delivery Network Works

1. Parcels from an out-of-town distribution centre are loaded onto a specially designed cargo tram (or into reserved space on a regular tram) at a *city-edge terminal*.
2. The tram carries the parcels along its scheduled route to *micro-depots* located at tram stops.
3. At each micro-depot, couriers on bicycles or on foot complete the *last-mile delivery* to individual addresses within a few hundred metres of the stop.

13.2 Public Transport as a Freight Carrier II



13.2 Public Transport as a Freight Carrier III

The book cites a study on cargo trams in Poland (Pietrzak and Pietrzak, 2021) as an example of this approach being tested in practice. The optimisation challenge is to schedule which parcels travel on which tram service, determine micro-depot locations, and plan the last-mile courier routes — all simultaneously. This is a complex multi-modal routing problem that the metaheuristics covered earlier in the book are well suited to tackle.

Key Insight

Public transport integration does not replace van delivery; it *concentrates* van activity on routes between the distribution centre and the city edge, while replacing the more polluting urban leg with a zero-marginal-emission tram service. The overall system becomes a two-tier logistics network, and optimising both tiers simultaneously is a rich research problem.

Python: Electric Vehicle Range Feasibility Check

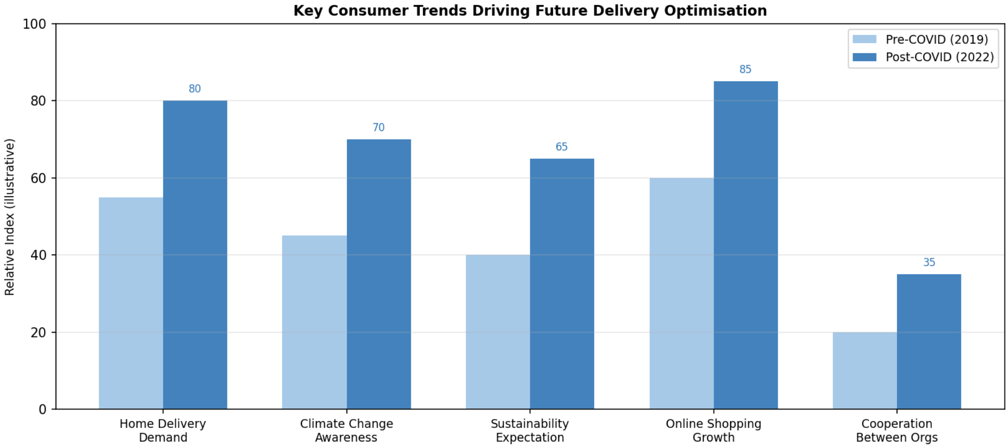
The following Python function illustrates how to check whether a proposed route is feasible for an electric vehicle given its battery capacity and the distances between stops. If the cumulative distance exceeds the range, the function identifies where the vehicle would need to stop for a recharge.

```python
def check_ev_feasibility(route, distance_matrix, ev_range_km,
                         charging_stations):
    """
    Checks whether a route is feasible for an electric vehicle.

    Parameters
    ----------
    route              : list of node indices, e.g. [0, 3, 7, 2, 0]
                         where 0 is the depot
    distance_matrix    : 2-D list / numpy array;
                         distance_matrix[i][j] = distance in km
    ev_range_km        : float, maximum range on a full charge
    charging_stations: set of node indices that are charging stations

    Returns
    -------
    dict with keys:
      'feasible'          : bool
      'charge_at'         : list of nodes where charging is needed
      'distance_segments': list of cumulative distances between charges
    """
    charge_at      = []
    segments       = []
```

13.3 Consumers — The Demand Side I

Technology on the supply side (better algorithms, autonomous vehicles, drones) is only one half of the picture. The other half is the *demand side*: what consumers expect from delivery services, and how those expectations are changing.

13.3 Consumers — The Demand Side II



13.3 Consumers — The Demand Side III

Three Overarching Trends Identified in the Book

1. **Increasing demand for home delivery.** The COVID-19 pandemic dramatically accelerated the long-running shift from in-store shopping to e-commerce. Home delivery volumes approximately doubled in many markets between 2019 and 2021, and this growth is expected to persist.
2. **Growing awareness of climate change.** Consumers are increasingly aware that every delivery has a carbon footprint, and a growing minority is willing to pay a premium for greener delivery options (e.g., consolidated next-day delivery rather than same-day).
3. **Pressure for sustainability from multiple directions.** Climate-change awareness is pushing both consumers and regulators to demand lower-emission delivery systems, creating a strong incentive for carriers to invest in electric fleets, route optimisation, and consolidation.

13.3 The Problem of Failed Deliveries I

One of the most costly and environmentally damaging outcomes in last-mile logistics is a **failed delivery**: the vehicle arrives at a customer address but nobody is home to accept the parcel. The book discusses this in detail because it illustrates the tight coupling between consumer behaviour and optimisation.

Why Failed Deliveries Are Expensive

- **Direct cost.** The vehicle has consumed fuel and driver time (or in the EV case, battery charge) without completing a delivery. The parcel must be returned to a depot and either re-delivered or collected.
- **Environmental cost.** The failed delivery represents "wasted" vehicle kilometres that produce CO_2 with no useful outcome.
- **Knock-on effect.** Returning to re-deliver disrupts the planned route for subsequent stops, potentially causing cascading late deliveries.

13.3 The Problem of Failed Deliveries II

Worked Example: Cost of a 10% Failure Rate

Consider a depot that dispatches 100 vehicles, each serving 50 customers per day. If the failed-delivery rate is 10%, then on average each vehicle fails to deliver at 5 stops. If re-delivery requires a dedicated return trip of 3 km per failed stop, and the vehicle emits 200 g CO_2 per km, then:

$$\text{Wasted distance per vehicle} = 5 \times 3 = 15 \text{ km}$$

$$\text{Extra emissions per vehicle} = 15 \times 200 = 3{,}000 \text{ g} = 3 \text{ kg } CO_2$$

$$\text{Fleet-wide extra emissions} = 100 \times 3 = 300 \text{ kg } CO_2 \text{ per day}$$

Over a year (250 working days) this amounts to $300 \times 250 = 75{,}000$ kg $= 75$ tonnes of avoidable CO_2. For a medium-sized carrier, this is a significant fraction of the fleet's total carbon footprint.

13.3 The Problem of Failed Deliveries III

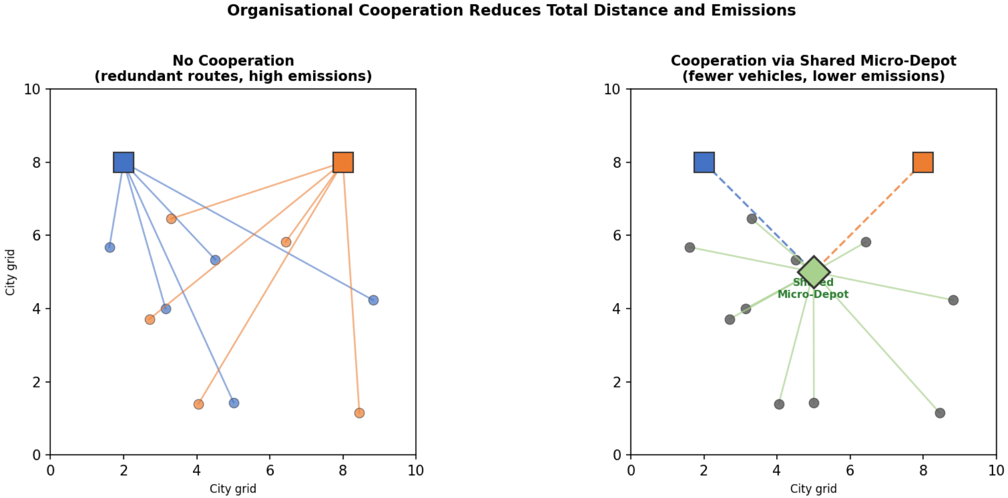
Key Insight

Reducing the failed-delivery rate is not just a customer-service problem; it is an environmental and economic one. Approaches include better delivery-window estimation, smart parcel lockers, and dynamic re-routing that responds to real-time customer availability data.

13.3 Cooperation Between Organisations I

A theme that runs throughout the book's conclusions is that many of the hardest challenges in delivery logistics cannot be solved by any single organisation acting alone. **Cross-organisational cooperation** offers significant efficiency gains, but requires trust, data sharing, and potentially new regulatory frameworks.

13.3 Cooperation Between Organisations II



13.3 Cooperation Between Organisations III

Forms of Cooperative Delivery

- **Shared micro-depots.** Two or more companies share a city-centre mini-warehouse. Parcels from different carriers are consolidated and delivered together by a single courier, halving the number of vehicles in the area.
- **Route sharing.** Carriers compare planned routes and identify overlapping segments. Where two vehicles are heading in the same direction, one can carry the other's parcels for part of the journey (analogous to ridesharing for freight).
- **Drone swarms across organisations.** Multiple companies' drone fleets share a city-level airspace management system that ensures collision avoidance and fair allocation of flight paths.

13.3 Cooperation Between Organisations IV

Worked Example: Savings from Cooperation

Studies cited in the book suggest that route-sharing between two medium-sized carriers typically reduces total distance driven by **15–25%** without increasing delivery times. For a combined fleet of 200 vehicles each driving 100 km/day, this amounts to:

$$\text{Distance saving} = 200 \times 100 \times 0.20 = 4{,}000 \text{ km/day}$$

At 200 g CO_2/km, this is 800 kg $=$ 0.8 tonnes of CO_2 saved *every single day*.

Key Insight

Cooperation transforms the optimisation problem from a single-company VRP into a *collaborative VRP* (ColVRP), where the objective function includes a fairness term to ensure each partner benefits. This is an active and largely unsolved research area.

13.3 Sustainability as a Design Constraint I

The book argues that sustainability — reducing carbon emissions, minimising wasted trips, and using vehicles efficiently — is no longer an optional add-on but a core design constraint for future delivery systems.

Consumer pressure, regulation (e.g., low-emission zones in European cities), and corporate net-zero commitments are all converging to make *green routing* the default mode of operation for forward-looking carriers.

Sustainability Levers Available to Carriers

Fleet electrification. Replacing diesel vans with EVs eliminates tailpipe emissions at the point of delivery. When charged on renewable energy, the lifecycle emissions approach zero.

Parcel consolidation. Combining multiple small orders into a single larger delivery (perhaps with a one-day delay) reduces the total number of vehicle trips. The book notes that even modest consolidation rates of 20–30% can cut vehicle-kilometres by 10–15%.

Green routing. Route optimisation algorithms can be extended to minimise fuel consumption or CO_2 rather than simply minimising distance or time. Because fuel consumption depends on speed, gradient, and load — not just distance — this requires more detailed vehicle models but is achievable with the techniques covered in this book.

Consumer-side interventions. Offering financial incentives for flexible delivery windows, consolidated deliveries, or parcel-locker collection can shift consumer behaviour in ways that make routing more efficient.

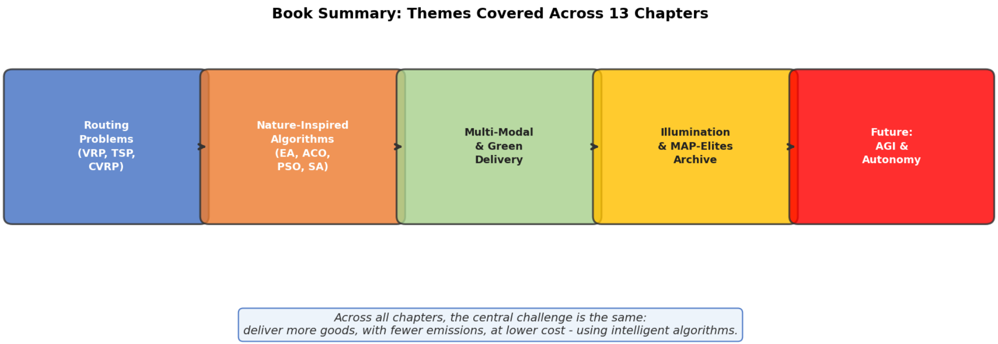
13.3 Sustainability as a Design Constraint III

Key Insight

The most powerful sustainability interventions often act on multiple levers simultaneously. For example, a carrier that electrifies its fleet *and* consolidates parcels *and* uses green routing can achieve far greater emissions reductions than any single measure alone. Multi-objective optimisation — a central theme of this book — is the natural mathematical framework for managing these trade-offs.

Book Summary: Thirteen Chapters in Perspective I

This concluding slide situates the chapter's forward-looking discussion within the broader arc of the book.



Book Summary: Thirteen Chapters in Perspective II

What the Book Has Shown

- **Chapters 1–4** established the fundamental routing problems (TSP, VRP, CVRP) and the first-generation nature-inspired algorithms (evolutionary computation, local search) that solve them.
- **Chapters 5–6** extended the single-objective framework to multi-objective problems and introduced the illumination paradigm (MAP-Elites), which produces a diverse archive of high-quality solutions rather than a single "best" answer.
- **Chapters 7–9** showed how to incorporate real-world data — geospatial information, traffic patterns, operational constraints — into the optimisation models, bridging the gap between academic benchmarks and practical deployment.
- **Chapters 10–12** applied these techniques to three concrete industries: food delivery, letter (postal) delivery, and parcel delivery. Each case study revealed industry-specific constraints and performance benchmarks.
- **Chapter 13** (this chapter) looks ahead to the algorithmic, technological, and societal changes that will define the next decade of delivery logistics.

Final Conclusions I

The book ends with a statement that captures the authors' central conviction:

Central Message

"The field of delivery optimisation stands at a moment of profound change. The algorithms developed over the past three decades — genetic algorithms, ant colony optimisation, simulated annealing, MAP-Elites — provide a solid mathematical foundation. But the problems ahead are harder, the systems are more complex, and the stakes are higher: climate change, urban congestion, and the expectations of billions of online shoppers are all converging on the logistics industry at the same time."

What the Next Generation of Solvers Must Do

- Handle heterogeneous, multi-modal fleets (vans, EVs, drones, trams) in a single model.
- Operate in real time as conditions change (dynamic VRP).

What This Book Provides

- A rigorous introduction to the mathematical structure of routing problems.
- A toolkit of nature-inspired algorithms that are both effective and adaptable.
- A framework (illumination / MAP-Elites) for

Final Conclusions II

Final Thought

Nature-inspired optimisation — algorithms that borrow the principles of evolution, ant foraging, and particle swarms — has proven to be remarkably well suited to the complex, noisy, constraint-rich problems of real-world delivery logistics. The challenge for the coming decade is to build these algorithms into intelligent, adaptive, sustainable systems that serve both the economy and the planet.

References I

The following works are cited in Chapter 13 of the book. They provide starting points for further reading on the topics covered in this chapter.

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